



High Ionic Conductivity Lithium Garnet Oxides of $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ Compositions

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Lithium garnet oxides of the composition series $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ ($x = 0-2$) were synthesized by the solid state reaction and characterized by the powder X-ray diffraction and the impedance spectroscopy. Single cubic phases were obtained between $x = 0.2$ and 2, while the end-member $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ exhibited a tetragonal phase. The lattice parameters of the cubic series followed the Vegard's law. The maximum bulk (9.6×10^{-4} S/cm) and total (6.9×10^{-4} S/cm) conductivities were achieved at $x = 0.3$ and $x = 0.2$, respectively at room temperature. Electrochemical tests with a hybrid solid/liquid electrolyte configuration were performed to evaluate the electrochemical performance of the solid electrolytes.

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There have been many studies devoted to the search of inorganic solid state electrolytes in substitution for liquid electrolytes in secondary lithium ion batteries.¹⁻³ Two of the most studied inorganic compounds are $\text{Li}_{1+x}\text{Ti}_{2-x}\text{Al}_x(\text{PO}_4)_3$ (LATP)^{4,5} glass ceramics based on the sodium super ionic conductor (NASICON) structure and perovskite oxides $\text{Li}_{3x}\text{La}_{2/3-x}\text{TiO}_3$ (LLTO).^{6,7} However, both LATP and LLTO contain Ti^{4+} which is susceptible to reduction on the anode side. For example, color changes were observed as these two compounds were in contact with lithium metal.^{6,8}

It was reported by Weppner et al.⁹ that cubic garnet oxides $\text{Li}_5\text{La}_3\text{M}_2\text{O}_{12}$ ($\text{M} = \text{Nb}$ and Ta) are lithium ionic conductors with conductivity values of around 10^{-6} S/cm at 25°C. The substitution of Zr over Nb/Ta enhanced the bulk conductivity to $\sim 5 \times 10^{-4}$ S/cm for $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$.¹⁰ This material was initially identified as a cubic phase with a space group of ($Ia\bar{3}d$, no. 230).¹⁰ It was recently found the cubic phase was possibly stabilized by the Al contamination from the crucibles, during the high temperature (>900°C) process.¹¹ The Al-free $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$, as well as $\text{Li}_7\text{La}_3\text{Hf}_2\text{O}_{12}$ and $\text{Li}_7\text{La}_3\text{Sn}_2\text{O}_{12}$, generally crystallizes in a tetragonal phase with a space group of ($I4_1/acd$, no. 142),¹¹⁻¹⁴ although Al-free cubic $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ powders were also reported, if they were calcined below 850°C.¹⁵ The tetragonal $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ phase was found to have a much lower bulk conductivity of 1.6×10^{-6} S/cm at 25°C.¹² Both $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ ⁹ and the "cubic" $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ ¹⁰ were found to be stable against molten lithium, which offers significantly better stability over Ti^{4+} containing LATP and LLTO. Recently, the lithium ion conduction in $\text{Li}_6\text{La}_3\text{ZrTaO}_{12}$ was reported.^{16,17} However, there have been no studies in the complete composition series $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$, which could have a combination of high conductivity and stability.

In this paper, the phase and ionic conduction behaviors of the $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ series ($x = 0-2$) were studied by the powder X-ray diffraction and impedance spectroscopy. An electrochemical cell was assembled with sintered solid electrolyte pellets to show the applicability of this material in lithium batteries.

Experimental

Polycrystalline powders with compositions $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ ($x = 0, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 1, 2$) (denoted as LLZTx thereafter) were synthesized by the solid state method. Starting materials of $\text{LiOH} \cdot \text{H}_2\text{O}$ (Alfa Aesar, 99.995% purity), La_2O_3 (Sigma Aldrich, 99.9% purity), ZrO_2 (Alfa Aesar, 99.7% purity), and Ta_2O_5 (Alfa Aesar, 99.85% purity) were ball-milled in 2-propanol for 24 hours with zirconia balls in polyethylene jars. 10 wt% excess lithium starting materials were added to compensate for the evaporation during calcinations and sintering. The mixture was then dried and heated in air

at 900°C for 10 hours in alumina crucibles with a heating rate 3°C/min. The ballmilling and heating were repeated once to enhance purity.

For the sintering of LLZTx series, around 0.5 atomic % of NaHCO_3 (J.T.Baker, 99.7-100.3%) and $\alpha\text{-Al}_2\text{O}_3$ (Alfa Aesar, 99.9%) were mixed with the obtained powders. Pellets were cold-pressed in a 1/2 inch die with a uniaxial press (Carver 3851) at 5 tons and then sintered in air at various temperatures in alumina crucibles for 36 hours. The heating rate was 3°C/min before 900°C and 1°C/min to the desired temperature, and the cooling rate was 4°C/min. For LLZTx compositions of $x = 0.2-0.5$, LLZT2, i.e. $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$, pellets were placed below the samples to prevent Al contamination from the crucible. All pellets were covered with powders of the same composition to minimize evaporations.

Powder X-ray diffraction (XRD) was used to characterize the phase of the calcined powders and sintered pellets with a Bruker D8 ADVANCE diffractometer using $\text{Cu-K}\alpha$ X-rays. The densities of the pellets were measured by the Archimedes' method in 2-propanol at room temperature. A scanning electron microscope (Carl Zeiss, EVO LS25) was used to check the microstructure of sintered pellets.

For electrical characterization, pellets were pasted with gold ink (PELCO) and cured at 700°C for 1 hour to improve adhesion, then assembled in Swagelok cells with stainless steel 316 rods as current collectors. Impedance measurements were performed in the frequency range 1 Hz to 500 kHz by a Bio-logic VMP 3 multichannel potentiostat, with a peak sinusoidal amplitude of 20 mV. A BINDER ED23 heating oven and a freezer were employed for variable temperature impedance tests.

To test the applicability of solid electrolytes for lithium ion batteries, a cell with a hybrid solid/liquid electrolyte was assembled. A sintered pellet of composition $\text{Li}_{6.7}\text{La}_3\text{Zr}_{1.7}\text{Ta}_{0.3}\text{O}_{12}$ was embedded in polyethylene (PE) and the whole disk was hermetically sandwiched by two liquid electrolyte compartments. The liquid electrolyte (LE) is 1 mol/L of LiPF_6 in dimethyl carbonate (DMC), dimethyl ethylene carbonate (DEC), and ethylene carbonate (EC) with a volume ratio of 1:1:1 (MTI). Lithium foil (Sigma-Aldrich, 0.38 mm thick) and LiFePO_4 composite film (MTI) were used as the anode and cathode, respectively. Both electrodes are 12 mm in diameter. Two pieces of polyethylene porous films (MTI) were employed as the separators. Stainless steel 316 rods were used as current collectors. Electrochemical performance test was carried out at room temperature at a constant current of 100 μA which corresponds to C/26 rate based on the active material of LiFePO_4 .

Results and Discussion

XRD characterization.— Fig. 1 shows the room temperature powder XRD patterns of LLZTx series calcined at 900°C. The end-member $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ exhibits a tetragonal phase, consistent with literature reports that used similar processing temperatures.^{11,12} The peak

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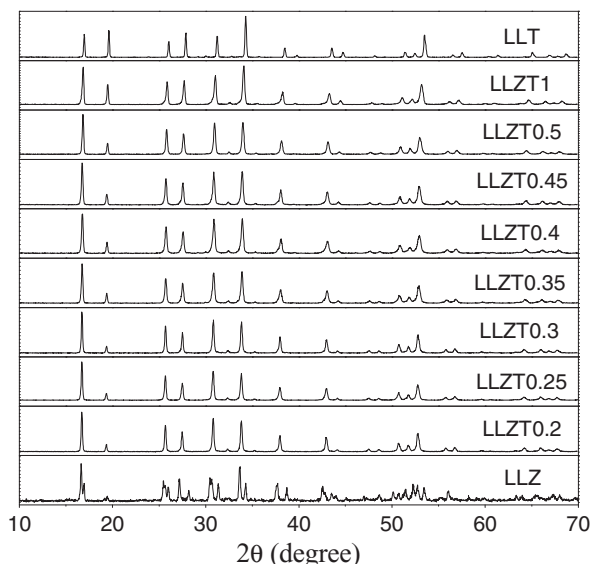


Figure 1. XRD patterns of powders calcined at 900°C of LLZTx or $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ series.

positions of all compositions except $x = 0$ match with those in the cubic $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ (PDF: 45-0110),^{18,19} The peaks shifted to higher angles as more Ta was added, suggesting a decrease of lattice parameters, since the ionic radius of Ta^{5+} (0.64 Å) is less than that of Zr^{4+} (0.72 Å) under the same six coordination.^{20,21} The Rietveld refinement with the General Structure Analysis System (GSAS) package^{22,23} was performed to obtain the lattice parameters of the cubic compositions. A structural model of space group of $1a\bar{3}d$, similar to that in Cussen et al.,¹⁹ was used as the initial structure. La adopts a Wyckoff position of 24c and Zr/Ta are randomly distributed on the 16a site. It is generally agreed that lithium disorder in garnet oxides has a complex mechanism and lithium location on the tetrahedral 24d, center octahedral 48g, and displaced octahedral 96h sites has been reported.²¹ Due to the low scattering factor of lithium to X-rays, lithium disorder was not the focus of the present XRD study; thus only 24d and 48g sites were adopted for Li in the refinement. The compositional dependence of lattice parameters is shown in Table I. The lattice parameter decreases with the increase of Ta content and a rough linear relation was seen in consistency with the Vegard's law.

Sintering behavior.— For the Zr-rich compositions ($x < 0.5$), the samples had stronger tendency to react with alumina crucibles at high sintering temperatures, based on the visual observation of sample color change from white to light brown. Lowering the sintering temperatures with sintering aids and application of $\text{Li}_5\text{La}_3\text{Ta}_2\text{O}_{12}$ as the buffer pellet minimized this reaction. The lowest sintering temperatures to obtain relative densities above 91% with almost zero open porosity were obtained for each chemical composition. They are summarized in Table I, along with the achieved density values.

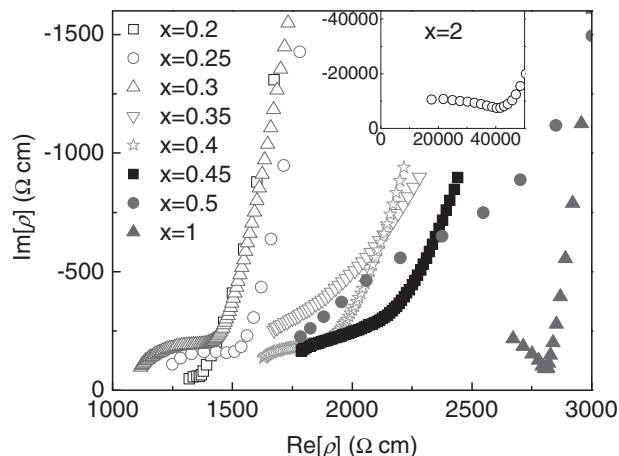


Figure 2. Nyquist impedance plots of $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ pellets at room temperature. The imaginary part of the resistivity is plotted against the real part.

The particle size in the sintered pellets is generally between 2 and 10 microns (e.g. the average particle size at $x = 0.3$ is 5 microns).

Electrical characterization.— Fig. 2 shows the Nyquist impedance plot of the LLZTx series from $x = 0.2$ to $x = 1$ measured at room temperature. Typical dimensions of the pellets are 10.8 mm in diameter and 3.5 mm thick. All resistance data were converted to resistivity for comparison. One arc was followed by a sloping line representing the blocking effect of gold paste to the ionic motion. The grain interior (bulk) and grain boundary resistivities were obtained by fitting the experiment data to a standard equivalent circuit $C_{\text{stray}} \parallel (L - R_{\text{GI}} - R_{\text{GB}} \parallel Q_{\text{GB}} - R_{\text{EL}} \parallel Q_{\text{EL}})$, with the help of Zview program.²⁴ The dash and parallel symbols represent the series and parallel connection, respectively. In the equivalent circuit, L and C_{stray} are the inductance and stray capacitance from the test leads, respectively. R_{GI} , R_{GB} , and R_{EL} represent the grain interior, grain boundary, and electrode resistance (resistivity), respectively; Q_{GB} and Q_{EL} are the constant phase elements (CPE) for the grain boundary and electrodes, respectively. The CPE impedance takes the form of $[(j\omega)^n Q]^{-1}$ where ω is the applied angular frequency and n is the parameter for the homogeneity of the conduction and dielectric responses. The normalized dielectric capacitance C can be calculated as $Q^{1/n} R^{1/n-1}$.²⁵ The measurements in Fig. 2 were performed with the calibrated cables and the stray capacitance was ignored. The inductance value was determined to be 2×10^{-7} H from measurements on standard resistors. The normalized capacitance values from the fitting are on the order of 10^{-9} F for the arcs in Fig. 2, which is consistent with the expected grain boundary contribution. The grain interior and total conductivity values are shown in Table I. The maximum room temperature grain interior conductivity is around 9.6×10^{-4} S/cm for $\text{Li}_{6.7}\text{La}_3\text{Zr}_{1.7}\text{Ta}_{0.3}\text{O}_{12}$, which is comparable to 9.5×10^{-4} S/cm reported for $\text{Li}_{6.75}\text{La}_3\text{Zr}_{1.75}\text{Nb}_{0.25}\text{O}_{12}$.²⁶ It is higher than $\sim 5 \times 10^{-4}$ S/cm reported¹⁰ for the “cubic” $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ and 4.3×10^{-5} S/cm reported²⁷ for $\text{Li}_6\text{La}_2\text{BaTa}_2\text{O}_{12}$. The conductivity

Table I. Lattice parameter, sintering temperature, relative density, and room-temperature grain interior and total conductivity of $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ series.

Composition (x)	0.2	0.25	0.3	0.35	0.4	0.45	0.5	1	2
Lattice parameter (Å)	12.9780	12.9801	12.9721	12.9591	12.9470	12.9489	12.9187	12.8900	12.8311
Temperature (°C)	1130	1130	1100	1130	1130	1130	1130	1100	1230
Relative density (%)	91.7	92.7	92.5	92.8	92.8	93.6	94.1	95.0	93.8
GI conductivity (10^{-4} S/cm)	7.8	8.8	9.6	7.7	7.2	6.9	6.7	4.4	1.6
Total conductivity (10^{-4} S/cm)	7.3	6.3	6.9	4.6	4.7	4.0	2.8	3.5	0.34

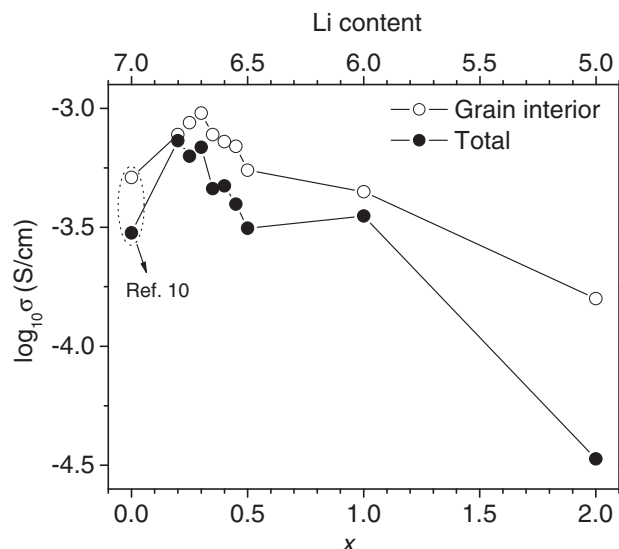


Figure 3. Composition dependence of room-temperature grain interior (bulk) and total conductivity of $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_{0.3}\text{O}_{12}$ series. For composition $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$, data reported from the reference 10 was used.

values from the present work are also plotted in Fig. 3 to illustrate their dependence on the lithium composition. As discussed before, a tetragonal phase was obtained for $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ ($x = 0$) in the present work so the literature data¹⁰ of the “cubic” phase were plotted. It is apparent that the peak grain interior was obtained at $x = 0.3$ while the peak total conductivity occurred at $x = 0.2$, due to the lower grain boundary resistance, which might be caused by the cation segregation and reaction with sintering aids. A peak conductivity was reported at $x = 0.25$ for the $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Nb}_x\text{O}_{12}$ series.²⁶ A qualitative explanation of the observed composition dependent conductivity will be presented here. The lithium ionic conductivity is the product of lithium ion concentration and mobility. From nuclear magnetic resonance studies, it is generally accepted that the octahedral sites in lithium garnet oxides take part in the conduction mechanism.^{21,28,29} It is expected that the mobility of a lithium ion on an octahedral site will be affected by the probability that the next octahedral jumping site is empty. Thus a maximum conductivity is expected at intermediate concentration values due to the trade-off between the carrier and vacancy concentrations on the octahedral sites. Variation of conductivities was observed from batch to batch. For example, the bulk conductivity values between 6.5 and 9.6×10^{-4} S/cm were obtained for $\text{Li}_{6.7}\text{La}_3\text{Zr}_{1.7}\text{Ta}_{0.3}\text{O}_{12}$.

The variable temperature Nyquist impedance plot on the LLZT0.3 pellet is shown in Fig. 4. Above 80°C , the grain boundary arcs can't be observed so Q_{GB} was removed from the equivalent circuit. Thus only the total resistance can be obtained. The inductance value was 3×10^{-6} H due to the extension test leads. The stray capacitance (4×10^{-11} F) was determined by performing the measurement on the same sample at room temperature with and without the test leads. This capacitance causes the kink in the high frequency data at 40°C in Fig. 4. The Arrhenius plot for the grain interior ($<80^\circ\text{C}$) and total conductivity values is shown in Fig. 5. Both the conductivity σ and product of conductivity and temperature σT have been used for the Arrhenius relation in the literatures. For the conductivity σ , the activation energies of LLZT0.3 are 0.37 eV (35.6 kJ/mol) and 0.36 eV (34.6 kJ/mol) for the bulk and total conductivity, respectively. For σT , the activation energies are 0.40 eV (38.5 kJ/mol) and 0.39 eV (37.6 kJ/mol) for the bulk and total conductivity, respectively. These values are comparable to those reported for the total conductivity σ of $\text{Li}_{6.75}\text{La}_3\text{Zr}_{1.75}\text{Nb}_{0.25}\text{O}_{12}$ (~ 30 kJ/mol)²⁶ or σT of $\text{Li}_6\text{La}_3\text{TaZrO}_{12}$ (0.42 eV).¹⁶

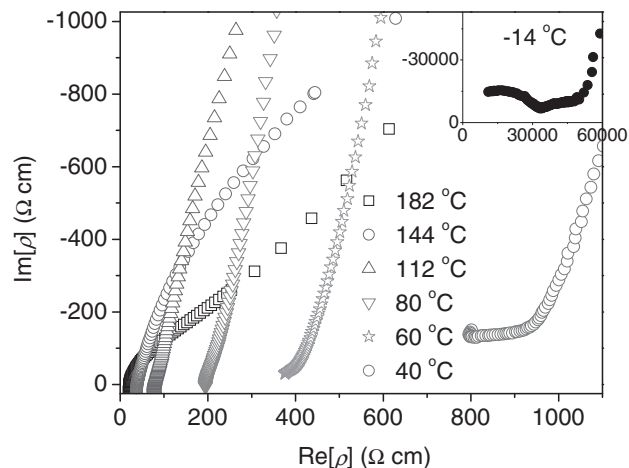


Figure 4. Nyquist impedance of variable impedance measurements of $\text{Li}_{6.7}\text{La}_3\text{Zr}_{1.7}\text{Ta}_{0.3}\text{O}_{12}$. The imaginary part of the resistivity is plotted against the real part.

Electrochemical test.— An electrochemical cell with a hybrid solid electrolyte (SE)-liquid electrolyte (LE) configuration shown in Fig. 6 was assembled and its electrochemical behaviors were studied. In this configuration, lithium ions have to move through the SE and SE/LE interfaces since the center disk is hermetically sealed against the liquid electrolyte. The charge/discharge curves of the first cycle are shown in Fig. 7a, along with the impedance change before and after cycling in Fig. 7b. The charge capacity was 154 mAh/g and discharge capacity was 153 mAh/g. In the Nyquist plot, the high frequency arcs of all spectra display resistance of between 800 and 900 Ohms, which is on the order of the expected value of the solid electrolyte disk (600 Ohms). Both the high frequency and low frequency features change with the cycling, which suggests complicated dynamics of the SE/LE interface and possibly mixed conduction of the solid electrolyte. More detailed studies of interfacial behavior of SE/LE interface and the mixed conduction of solid electrolytes are in progress. This experiment provides a demonstration that lithium

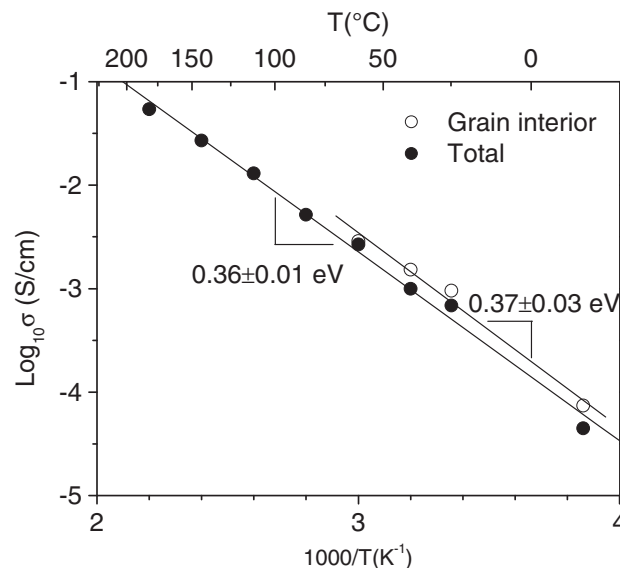


Figure 5. Arrhenius plot for the grain interior and total conductivity of $\text{Li}_{6.7}\text{La}_3\text{Zr}_{1.7}\text{Ta}_{0.3}\text{O}_{12}$. The straight lines represent the fitting of Arrhenius relation with conductivity values.

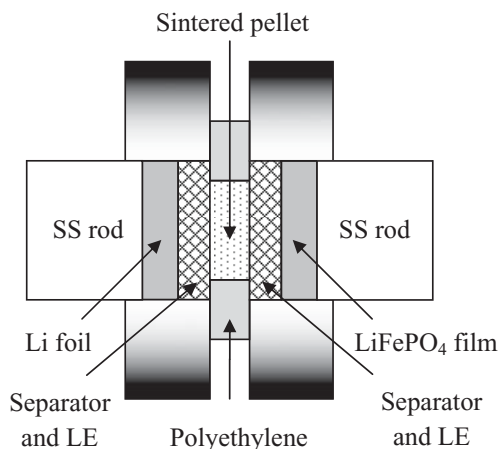


Figure 6. Schematic of the electrochemical cell with a sintered $\text{Li}_{6.7}\text{La}_3\text{Zr}_{1.7}\text{Ta}_{0.3}\text{O}_{12}$ pellet. SS: stainless steel 316; LE: liquid electrolyte; PE: polyethylene.

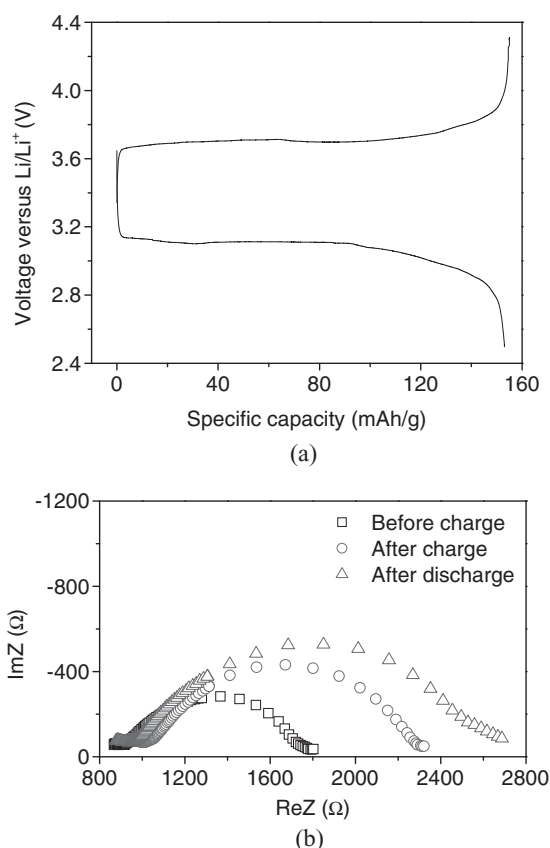


Figure 7. (a) Charge/discharge curves of the electrochemical cell with solid electrolyte at room temperature. The charge capacity was 154 mAh/g and discharge capacity was 153 mAh/g. (b) Impedance evolution of the electrochemical cell in the first cycle.

garnet oxides support ionic transport that is relevant for lithium battery applications.

Conclusions

Lithium garnet oxides with formulas $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ ($x = 0-2$) were successfully synthesized by the solid state reaction method. Single cubic phases were observed in the composition x range between 0.2 and 2, while $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ ($x = 0$) displayed a tetragonal structure. The lattice parameters decreased with the addition of Ta due to the smaller ionic radius of Ta^{5+} compared with Zr^{4+} , following the Vegard's law. Pellets with relative density above 91% were obtained from high temperature processing with sintering aids. The maximum bulk conductivity was achieved for $x = 0.3$, with a value of 9.6×10^{-4} S/cm at room temperature and an activation energy of 0.37 eV. The total conductivity of the $x = 0.3$ composition is 6.9×10^{-4} S/cm at room temperature with an activation energy of 0.36 eV. The maximum total conductivity in $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Ta}_x\text{O}_{12}$ was achieved for $x = 0.2$, due to the lower grain boundary resistance. These lithium garnet oxides exhibit lithium ionic transport that is relevant for lithium battery application.

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